Mark Vasile

Applied AI / Machine Learning Engineer

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SUMMARY

Applied AI and machine learning engineer with 25+ years of software engineering, from Linux networking and C/C++ to LLM pipelines in Rust and Python. I've created Kawa Code AI as a personal project, an AI Decision Manager that captures the reasoning behind code changes and serves it to AI coding agents over MCP. I've also fine-tuned and evaluated open-weight LLMs, designed multi-model LLM systems with explicit cost/quality tiering, and built neural networks from first principles, from a GPT-style decoder in PyTorch to an MNIST classifier in TypeScript. Inventor on two granted patents (US, Japan). Fluent in English, Japanese and Romanian.

TECHNICAL SKILLS

Machine learning: PyTorch, transformer / attention internals, MLPs and backpropagation, LoRA / QLoRA fine-tuning (MLX-LM), 4-bit quantized inference, synthetic data generation, Qwen, Gemma, Llama

LLM systems: Claude (Opus, Sonnet, Haiku) and Anthropic API, MCP servers, Claude Code hooks, multi-agent orchestration, LLM-as-judge, RAG with Voyage embeddings, knowledge graphs, structured JSON output, Ollama, llama.cpp, Cursor, LM Studio

Languages: Python, Rust, TypeScript, Go, C/C++, PHP, Haskell, SQL

Platforms: Rust (axum / tokio), Tauri, Vue, React, Node.js, MongoDB, PostgreSQL, Redis, Kafka, AWS, GCP, Kubernetes, Docker, Linux administration and networking

MACHINE LEARNING PROJECTS

- **GPT-style decoder from scratch (Python, PyTorch).** Implemented a small trainer+decoder transformer (embeddings, causal multi-head self-attention, training loop and sampling) to study how attention works and the role of entropy in training loss and generation.
- **Neural network for image recognition (TypeScript, 2023).** Built a multilayer perceptron with two dense weight matrices on mathJS, with hand-written forward pass and backpropagation, trained and tested on the MNIST handwritten-digit dataset.

EXPERIENCE

Founder & AI Engineer | Kawa Code (kawacode.ai) Apr 2026 – Present

AI Decision Manager: auto-captures developer intents and decisions as curated context and transparently surfaces them to AI coding agents (Claude Code, Cursor). Sole founder; designed and built the full system, ~190k lines of Rust and TypeScript.

- **Fine-tuned Qwen3-14B** (4-bit, LoRA via MLX-LM) to turn developer intents and decisions into structured JSON on current-production-shaped data.
- Traced the hallucinations to the data, not the training. Rebuilt the dataset so every label is computed from its input, and generated approx. 1,300 synthetic examples in which Claude writes only inputs and code computes labels, seeded from the training split to prevent validation leakage.
- Used the model as a local MLX inference service in Python (~30 tok/s on Apple M5).
- **Built a multi-stage LLM pipeline in Rust** that mines git history into intents and decisions: 120 commits became 49 curated decisions in 15 minutes for $2.
- **Designed an AI conflict resolver for concurrent coding agents:** file-set locks, an N-way semantic LLM judge over full files, an adversarial verification pass, and a four-tier gate that auto-applies only the safe tier, inside agent-owned git worktrees.
- Built completion-time distillation and conflict detection: Sonnet condenses an agent's in-flight reasoning into durable decisions, and a Haiku judge checks each one against the repository's existing decisions, blocking contradictions before they persist.
- Implemented retrieval in a Rust API: Voyage code embeddings and semantic search over decisions, plus a Relate Engine that uses LLMs to extract key points from prose and build a multi-axis relationship graph.
- Published the @kawacode/mcp server (20+ tools) for Claude Code
- Built an admin dashboard and telemetry that weighs the context tokens injected per turn against value delivered: decisions captured, conflicts prevented and acted-on rate.

Creator | Code Awareness dev-tool Jan 2022 – Mar 2026

Low-noise collaboration developer tools; the foundation that later became Kawa Code AI.

- Built a VS Code extension that highlight the lines teammates are changing in their uncommitted working copies, so conflicts surface before merge, without detracting focus by showing actual real-time multi-cursor collaboration UX. Implemented the technique patented in JP 7150002. Extended to Emacs, Vim and JetBrains over Unix sockets and Windows named pipes.
- Built the agent coordinator: per-session identities for AI coding agents, intent tracking, and detection of overlapping work across concurrent agents and human teammates. Zero-knowledge handling: diffs are encrypted client-side, the server never sees actual code.
- Built an i18n extension for multi-lingual dev teams, an AST-based code translation extension (TypeScript Compiler API + Claude) that renders identifiers and comments in the developer's native language.

Staff Engineer + Sr Manager | Signify Health May 2024 – Apr 2026

- **Led AI-assisted engineering for two teams** (application + data engineering) using Cursor with Claude Sonnet and Opus: writing and reviewing thousands of lines of code a day, producing designs and plans from Jira and architecture documents, and analyzing meeting notes.
- Unified ~25 disparate QA processes into one system driven by AI-generated test data that mirrors production patterns, orchestrating interconnected team pipelines for end-to-end validation and alerting.
- Architected an ETL pipeline that automated the year-end rollover, improving throughput ~100x and replacing a 3-week multi-team manual cleanup with a single-day mostly-automated verification.
- Built a visual, diagram-based rules engine using Camunda, which replaced brittle SQL stored procedures for thousands of business rules; set up a prod-shadow environment validating pipelines, while keeping anonymized production data under HIPAA.

Tech Lead & Systems Architect | Pixelberry Studios Feb 2022 – Aug 2023

- Architected an enterprise multi-cultural / multi-lingual CMS (React, Express, DynamoDB, AWS) with geo-scheduled releases, a unified catalog and admin interface and multi-market advertising; this replaced a fragmented operator network across global markets.
- Led and mentored a 4-engineer team; instituted weekly pair programming that caught bugs before review and produced the highest team morale in the organization.

Front-end Developer | Rakuten Japan Oct 2021 – Jan 2022

- Built admin dashboards and reporting tools for Rakuten's network-wide eSIM platform in a multicultural, distributed team.

Team Lead | XICA Japan Jun 2018 – Jun 2020

- Led web and data initiatives for a marketing-analytics product built on linear regression models; decomposed the data-processing monolith into independent pipelines and coached a 4-person team in regression analysis, vueJS, D3 visualization, and micro-services.
- Rebuilt ~200 files of plain HTML/JS into a TypeScript + Vue architecture; mentored a direct report from analytics algorithms into API development and promotion.

Senior Web Developer | Boston Scientific · Compass Professional Health Services · TailLight May 2010 – Mar 2017

- Delivered production web and mobile applications across medical-device, healthcare-services and startup environments. NodeJS, PHP, Java, PostgreSQL.
- Earlier career: network engineering, Linux administration, and C/C++ and PHP development.

PATENTS

- **JP 7150002 B2** — Low noise collaboration system. Inventor, granted 2022.
- **US 10,454,599 B2** — Visualization of electromagnetic wave distribution and collaborative web-based design for optimal emitter placement. Co-inventor, granted 2019.

EDUCATION & CERTIFICATIONS

B.S. Quality Management and Commerce — Academy of Economic Studies, Bucharest, Romania

MIT — Architecture of Complex Systems · NSF I-Corps · MongoDB Certified DBA (10gen)